KanaMain — Project Brief

Oct 29, 2026 · @Stuart

Purpose and goals

KanaMain is a web app that teaches beginners to read hiragana and katakana, built as a full-stack TypeScript portfolio project that shows employers Stuart can design, build and deploy a complete product.

- **Users:** learners, called *students* in this brief, who study kana by row and then test themselves in a timed typing game.
- **Status:** a personal project with no link to Deakin University, built as a learning exercise with Claude as a coding partner.
- **Portfolio test:** a visitor to the GitHub repo should find a live demo, a clear README, a clean commit history, tests, a React and TypeScript front end and, in Phase 2, a SQL back end.
- **Audience:** not marketed to the public, so correctness and polish matter more than scale.

Scope

Version 1 covers the 214 standard hiragana and katakana cards. Kanji is not taught.

In scope

- Hiragana and katakana, each with three variants: monographs (46 cards), diacritics (25) and digraphs (36).
- A study page with a short introduction to each script, a typing game, personal bests and achievement badges.
- Basic spoken audio through the browser's speech synthesis.

Out of scope for now

- Teaching kanji. A few kanji appear only as badge decoration, always with furigana and an English meaning.
- Vocabulary lessons and sentences. Version 1 uses no words at all.
- Public marketing, social features and leaderboards.

Screens and navigation

Seven screens, reached from a deliberately simple home menu with the KanaMain logo above it.

Screen	What it does
Home	Logo plus a four-item menu: Play now, Study, Stats, About
Study	Browse every kana by row, filter, mark rows complete, hear each sound
Play setup	Pick writing system and variants for a run
Play	The typing game: one card at a time
Results	Time, cards reviewed, first-guess accuracy, missed kana
Stats	Best time and best accuracy per category, plus badges
About	What the app is and who built it

Study page

The study page lets a student see every kana, grouped so they can track what they have covered.

- **Layout by row:** each row has its own section, for example A I U E O, then KA KI KU KE KO, and the same for diacritic and digraph rows.
- **Filter:** by writing system (hiragana or katakana) and by variant, so a student can study one area at a time.
- **Mark complete:** a student can mark any row as complete once they feel they have mastered it. This is a personal study tracker and is kept completely separate from game results.
- **Cards:** each card shows the kana with its romaji underneath. There is no separate detail screen.
- **Audio:** tapping a card speaks the kana using the browser's built-in Japanese speech synthesis. Cards stay silent if the device has no Japanese voice. Audio sits behind one hook, so better recordings can be swapped in later without touching the screens.

Script introductions. A short paragraph at the top of the Study page tells students why each script exists, shown for whichever script is selected. It covers where the script came from and what it is used for today. Individual kana have no notes in version 1; teaching each kana in depth is left to a later Lessons feature.

Game rules

A run shows one kana card at a time; the student types its romaji and the game rewards first-guess accuracy.

1. **Answering:** a correct answer (any accepted spelling, such as shi or si) advances the game automatically.
2. **Wrong answers:** a wrong answer is only judged when the student presses Enter. The card shakes and the student can try again as many times as they like.
3. **Flipping:** if the student is stuck, clicking the card flips it to reveal the romaji. A flipped card counts as a miss and does not come back later in the run.
4. **Streak:** the streak counts correct first guesses in a row. A wrong submission or a flip ends it. The results show the best streak of the run.
5. **Timer:** starts when the student submits their first answer, right or wrong, and stops on the last answer.
6. **Stopping early:** a student can end a run at any point and still see results, but a stopped run never counts toward personal bests or badges.
7. **Results:** time taken, number of cards reviewed, percentage correct on the first guess, and the missed kana highlighted.

An *error* for accuracy and badges means a wrong answer submitted with Enter or a flipped card. Letters corrected before pressing Enter do not count, which protects mobile typists from slips.

Filters and personal-best categories

Students choose a writing system and then one, two or all three variants, which gives 21 categories; every run uses the full card set for that choice, with no card-count picker.

- **Writing system:** Hiragana, Katakana or Both.
- **Variants:** Monographs, Diacritics and Digraphs, in any combination, with All as a shortcut.
- **Personal bests:** kept per category, so times are only compared between runs of the same size.

Variants selected	One script	Both scripts
Monographs	46	92
Diacritics	25	50
Digraphs	36	72
Monographs + diacritics	71	142
Monographs + digraphs	82	164
Diacritics + digraphs	61	122
All three	107	214

The card counts above are cards per run. In the data, the group currently called "basic" is renamed "monographs" to match this wording.

Later versions. Small っ/ッ, the long-vowel mark ー and extended katakana such as ファ and ティ are not in version 1. They are best learned alongside words, and this app teaches single symbols. They return if the app grows to teach words and sentences.

Stats and badges

The Stats page shows two numbers per category, best time and best accuracy, plus 11 badges; there is no run history and no reset button.

- **Best time:** completed runs only.
- **Best accuracy:** the highest percentage of cards answered correctly on the first guess.
- **Earning a badge:** finish a run with the exact filter selection the badge names, with 100% first-guess accuracy: no wrong submissions and no flips. A larger run does not award the smaller badges.
- **Locked badges:** shown greyed out with まだまだです ("not there yet").

Each badge shows one kanji word with furigana above the kanji, so students can read it and learn it. A script mark sits in front: 平 for hiragana, 片 for katakana. The okurigana (the kana after the kanji, such as む) stay in hiragana, and the furigana is always in hiragana, as is conventional; it covers only the kanji itself. The Stats page also carries a short plain-English explanation of what kanji and furigana are, and why the badges use them.

# (hiragana / katakana)	Selection	Badge	Furigana	Meaning
1 / 6	Monographs	初め	はじ	begin

# (hiragana / katakana)	Selection	Badge	Furigana	Meaning
2 / 7	Diacritics	進む	すす	advance
3 / 8	Digraphs	伸びる	の	extend, grow
4 / 9	Monographs + diacritics	重ねる	かさ	layer, combine
5 / 10	All three variants	流れ	なが	flow
11	Both scripts, all variants	達人	たつじん	master

The master badge 達人 (script mark 仮名) is earned by a single flawless run of all 214 cards. The two combinations without a badge, monographs + digraphs and diacritics + digraphs, still have personal bests.

Look, devices and accessibility

The app is designed for phones first, iPads second and desktops third, and must work on all of them.

- **Palette:** navy, white and red, taken from the logo. Sampled from the logo files, the navy is about `#1a2e66` and the red about `#bc002d`; both are approximate because the files are JPEGs.
- **Logo:** two versions supplied, a square mark (red circle with あ, "KanaMain" in navy and red, カナマイン beneath) and a wide navy banner. Both will be rebuilt as SVG so they stay sharp at any size and can double as the browser-tab icon.
- **Badge colours:** hiragana badges in red, katakana badges in navy, and the master badge in both. Colour is only a reinforcement; the 平 and 片 marks carry the meaning.
- **Reduced motion:** the card shake and flip are replaced with a simple colour flash and fade for students who prefer reduced motion.
- **Other accessibility:** fully keyboard operable, screen-reader labels on cards and badges, readable contrast, and no information carried by colour alone.

Typography. Three Google Fonts, one per role, each set as a CSS variable so it can be swapped later.

Role	Font	Where it is used
Kana	Kosugi	The large kana on study and game cards, and any Japanese labels
Kanji badges	Zen Old Mincho	The badge kanji

Role	Font	Where it is used
English	Roboto	All English text, menus and buttons

The font stack lists Roboto first, then Kosugi, then the device's own Japanese font, so English letters and kana each get the right font automatically. Furigana on the badges uses Kosugi at a readable size, because thin mincho strokes are hard to read when small. Each Japanese font is served as a small subset covering only the kana and the badge kanji, to keep phones fast.

Technical approach

The project is TypeScript end to end, built front end first, with the storage and audio kept behind small interfaces so Phase 2 can plug in without rewriting screens.

Area	Choice
Language	TypeScript with strict mode on
Front end	Vite, React and React Router
Game logic	A typed state machine (waiting, wrong and retrying, revealed, finished) that replaces the old timer-based code
Data	The existing 214-card `kana.json`, typed, with "basic" renamed "monographs"
Storage (Phase 1)	Browser storage behind a storage interface for progress, bests and badges
Audio	Browser speech synthesis (Japanese voice) behind one audio hook
Testing and CI	Unit tests for game logic, scoring and badge rules, run on every push by GitHub Actions
Source control	GitHub
Hosting	Netlify, deploying from the GitHub repo, with a redirect rule so page refreshes work on client-side routes
Back end (Phase 2)	Node, Express and TypeScript with a PostgreSQL (SQL) database, JWT authentication and hashed passwords, hosted separately from the front end

Area	Choice
Accounts (Phase 2)	Guest mode keeps working; signing in merges the guest's local progress into the account

The tools above are proposals, to be confirmed when the project is scaffolded.

Milestones

The build runs in two phases, each ending in a working deployed app; badges are built after the core and are the first thing cut if time runs short.

Phase 1: front end (gate: live on Netlify)

1. **Foundation:** Vite, React and TypeScript repo on GitHub, CI, Netlify deploy, SVG logo, palette and routing shell.
2. **Data and Home:** typed kana data and the four-item home menu.
3. **Study page:** rows, filters, mark complete, audio and the script introductions.
4. **Game:** play setup, state machine, shake and flip, results screen.
5. **Stats and badges:** personal bests for the 21 categories and the 11 badges.
6. **Polish:** accessibility and reduced-motion pass, phone and iPad checks, tests, README.

Phase 2: back end (gate: sign-in works end to end)

1. **API and schema:** Express API and SQL tables for users, bests, completed rows and badges.
2. **Accounts:** sign-up, sign-in and guest-to-account merge.
3. **Deploy:** host the API and database, connect the front end, update the README.

Capacity is two full project days a week, with extra time on some weekends, alongside three university days. Milestone dates will be set once the foundation milestone shows the real pace.

Later: Lessons (idea, not in version 1). A guided lesson for each row (vowels, K row, S row and so on): a short explanation, tracing each character, a tap-the-sound quiz with four options, a build-the-word exercise using the row's own kana (for example あい), a memory check and a final matching exercise. It adds word-level content, stroke-order data and several new exercise types, so it is parked until Phase 1 is live. Phase 1 keeps the door open by organising data by row, building game logic as separate exercise components and keeping audio behind one hook.

Open items

No decisions are open for version 1.

Still to supply: the logo files in a larger size, or the originals, so they can be rebuilt as SVG.